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30 APRIL 1954
SUPERSEDING
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27 March 1952

FEDERAL SPECIFICATION

CLEANING COMPOUND, BREAD BAKING PAN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers bread baking pan cleaning compound for use in removing carbonaceous and greasy soil from both iron and tinned baking pans. This compound is not intended for use on sweet-goods bakepans.

2. APPLICABLE SPECIFICATIONS, STANDARD, AND OTHER PUBLICATION

2.1 The following specifications and standard, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

P–S–536—Soap and Soap-Products; (Including Synthetic Detergents); Methods for Sampling and Testing.

QQ–I–706—Iron and Steel; Sheet, Tinned (Tin-Plate).

RR–S–366—Sieves; Standard, Testing.

UU–S–48—Sacks, Paper, Shipping.

Military Standard:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under "General Information" in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.

(Copies of Military specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification.

Consolidated Classification Committee:

Consolidated Freight Classification Rules.

(Application for copies of the above mentioned publication should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

2.3 Specifications and standards applicable only to individual departments are listed in section 7.

3. REQUIREMENTS

3.1 Material.—Baking pan cleaning compound shall be a mixture of sodium carbonate, sodium metasilicate, and sodium chromate in any free flowing form such as powder, granular, crystalline, or flake. Abrasives or inert fillers shall not be present.

3.2 Moisture.—Moisture content shall not exceed 25 percent by weight when determined as specified in 4.4.1.

3.3 Carbonates.—Carbonates, calculated as CO_2, shall be between 17 and 21 percent by weight when determined as specified in 4.4.2.

3.4 Total (active) alkalinity.—Active alkalinity, calculated as Na_2O, shall be between 24 and 28 percent by weight when determined as specified in 4.4.3.

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3.5 **Silicates.**—Silicates, calculated as SiO_2, shall be between 11 and 15 percent when determined as specified in 4.4.4.

3.6 **Chromates.**—Chromates, calculated as CrO_3, shall be between 3 and 4 percent by weight when determined as specified in 4.4.5.

3.7 **Solubility.**—Matter insoluble in distilled water at a temperature of 100° Centigrade shall not exceed 0.50 percent by weight when determined as specified in 4.4.6.

3.8 **Corrosion.**—When tested as specified in 4.4.7 the compound shall not cause etching, discoloration, or spangling of tin-plated specimens. The weight loss shall not exceed one-half milligram per test strip (2½ inch × ⅝ inch).

3.9 **Coarse particles.**—Not more than 5 percent by weight of the compound shall be retained on a No. 8 sieve when tested as specified in 4.4.8.

3.10 **Labeling.**—Each container of baking pan cleaning compound shall contain a card durably and legibly marked with the following information:

Directions for use:

(1) *Equipment.*—For cleaning a large number of pans, a metal tank provided with wash and rinse compartments, closed steam coil, continuous overflow device, and cover is most suitable. A sink, basin, or any improvised equipment may be used for cleaning a small number of pans.

(2) *Amount of cleaner.*—Use 2 to 4 ounces per gallon of water depending on the degree of contamination.

(3) *Temperature of solution.*—Not less than 185° Fahrenheit and preferably boiling.

(4) *Cleaning operation.*—Immerse the pans, back to back, in the hot solution so that they are completely covered. Boil for 4 to 8 hours or until the soil is thoroughly softened. Remove the pans, rinse thoroughly with hot water, brush loose soil with a stiff bristle or fiber brush, and finally rinse again with hot water. Dry the pans, preferably in an oven, or by wiping thoroughly with clean dry cloths. Cleaned pans should be stored in accordance with approved methods.

(5) CAUTION: Keep hands and arms out of solution and avoid splashing to prevent injury to the skin. Use an iron hook for removing the pans from the cleaning solution. Personnel should wear rubber gloves and aprons and be provided with goggles for protection to the eyes.

Note.—It is important that the pans be thoroughly and completely rinsed with hot water after removal from the cleaning solution.

3.11 **Workmanship.**—The component ingredients shall be intimately assembled and processed in accordance with the best commercial practice for manufacture of this type of material. Baking pan cleaning compound shall be thoroughly mixed and present no evidence of segregation of individual constituents, or of lumping or caking. The product shall be stable in a sealed container and not subject to abnormal change with age.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.

4.1.1 *Lot.*—For purposes of sampling, a lot shall consist of not more than 60,000 pounds of material offered for inspection at one time. In case material is produced by a continuous-run process, the lot shall contain material from only one continuous-run. Material in the inspection lot shall be identified by order of production (in the case of a continuous-run process) or by batch number (in the case of a batch process) until ultimate action is taken by the Government inspector as to the acceptance or rejection of the lot.

4.1.2 *Sampling procedure.*—From each inspection lot, the Government inspector shall take three separate 1 pound samples. In case the material is produced by a batch process and the inspection lot contains more than two batches, the three samples shall normally be taken from different batches, from time to time; however, at the discretion of the inspector, two or three of the samples shall be taken from the same batch, in which case the samples shall be

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obtained in a manner calculated to disclose any nonuniformity of the material within the batch. Where material is produced by a continuous-run process, the three samples shall be taken so as to represent respectively, the first part, the middle part, and the last part of the run which produced the inspection lot. Each sample shall be thoroughly mixed and divided into three equal portions. The portions shall be placed in separate, clean, dry metal or glass containers which shall be sealed and carefully marked. One of the portions of each sample shall be forwarded to a Government laboratory designated by the procuring activity, one shall be delivered to the contractor, and one held by the Government inspector to be used for retests in case of dispute.

4.1.3 *Sampling for inspection of containers.*—A random sample of filled containers shall be taken by the Government inspector in accordance with Military Standard MIL-STD-105 at Inspection Level I and Acceptable Quality Level = 2.5 percent defective to verify compliance with all requirements regarding fill, closure, marking and other requirements not requiring tests.

4.2 Inspection.

4.2.1 *Inspection of containers.*—Each sample filled container shall be examined for defects of construction of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each filled container shall also be weighed to determine the amount of contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL-STD-105 the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for Government inspection provided the contractor has removed or repaired all nonconforming containers.

4.3 Lot acceptance tests.

4.3.1 *Procedure.*—The samples selected in accordance with 4.1.2 shall be subjected to the tests specified in 4.4. If any sample is found not to be in conformance with this specification (see 4.3.2) the entire lot which it represents, if produced by continuous-run process, shall be rejected; or if the lot is produced in batches, all untested batches in the lot shall be rejected together with the batch from which the sample was selected.

4.3.2 *Retests.*—If, in the opinion of the contractor, failure of a sample subjected to inspection tests at a Government laboratory to conform to any one of the requirements of this specification was due to faulty test methods, the test may be repeated on the contractor's retained portion of this sample at the contractor's laboratory. In case of the inconsistencies of results, the tests shall be repeated on the Government-retained portion of a sample. If the Government-retained portion of a sample is found to be nonconforming, the entire sample shall be finally judged nonconforming.

4.3.3 *Action in case of rejection.*—Rejected material may be resubmitted for Government inspection and testing provided the manufacturer, after having been informed of the reasons for rejection, has reworked the material or has tested it thoroughly for the deficiency noted and has removed all nonconforming material. In the case of rejected material produced by a batch process, such tests shall be made on each batch.

4.4 Test methods.

4.4.1 *Moisture.*—Determine moisture by method 20.2 of Federal Specification P-S-536.

4.4.2 *Carbonates.*—Determine carbonates as CO_2 by method 140.1 of Federal Specification P-S-536.

4.4.3 *Total (active) alkalinity (as Na_2O).*

4.4.3.1 *Reagents.*

4.4.3.1.1 *Standardized hydrochloric acid (0.25N).*—Accurately prepare and standardize a solution of hydrochloric acid.

4.4.3.1.2 *Phenolphthalein indicator.*—Dissolve one gram of phenolphthalein powder in 50 ml. of 95 percent ethyl alcohol and dilute the resulting solution with 50 ml. of distilled water.

4.4.3.2 *Procedure.*—Accurately weigh a one-gram sample into a 300-milliliter Erlenmeyer flask. Add 100 ml. of distilled water, 3 drops of phenolphthalein indicator and titrate with 0.25 normal hydrochloric acid (HCl) to a faint pink end-point that lasts for

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30 seconds. Active alkalinity as Na_2O is calculated as follows:

$$\frac{A \times B \times 3.1}{C} = \text{percent } Na_2O$$

where:

A = ml. of HCl
B = normality of HCl
C = weight of sample

4.4.4 *Silicates (as SiO_2).*—Accurately weigh a 1-gram sample into a 250-ml. beaker. Dissolve sample in 50 ml. of distilled water, add 25 ml of concentrated hydrochloric acid, and evaporate to dryness on a steam bath. Bake the residue for 1 hour at 105° to 110° C. Add 5 ml. of concentrated hydrochloric acid, 50 ml. of distilled water, and heat on the steam bath to dissolve soluble salts. Filter through a No. 40 Whatman filter paper and wash the residue thoroughly with hot dilute (1:20) hydrochloric acid, transferring the residue quantitatively to the filter. Ignite the filter paper and silica residue to constant weight at approximately 1000° C., weigh, and calculate the percentage of silica in the sample.

4.4.5 *Chromates (as CrO_3).*

4.4.5.1 *Reagents.*

4.4.5.1.1 *Standardized potassium permanganate (0.03N).*—Dissolve 0.95 gram $KMnO_4$ in 1000 ml. distilled water. Boil for 5 to 10 minutes. Allow to stand for at least a week. Filter through asbestos into a clean, glass-stoppered bottle. Standardize against Bureau of Standards sodium oxalate.

4.4.5.1.2 *Ferrous ammonium sulfate (0.05N).*—Dissolve 12 grams of

$$Fe(NH_4)_2(SO_4)_2 \cdot 6H_2O$$

in a mixture of 950 ml. of distilled water and 50 ml. of concentrated sulfuric acid.

4.4.5.1.3 *Orthophenanthroline indicator.*—Dissolve 1 gram of orthophenanthroline monohydrate in a few ml. of distilled water containing 0.468 gram of ferrous sulfate ($FeSO_4 \cdot 7H_2O$). Dilute to 68 ml. with distilled water.

4.4.5.2 *Procedure.*—Accurately weigh a 0.5-gram sample into a 300-ml Erlenmeyer flask. Dissolve the sample in 100 ml of distilled water, add 3 ml of concentrated sulfuric acid and exactly 25 ml of ferrous ammonium sulfate solution. Add 3 drops of orthophenanthroline indicator and titrate the excess ferrous ammonium sulfate with standardized potassium permanganate. In a like manner conduct a blank titration using exactly 25 ml of the ferrous ammonium sulfate solution. Chromate, as CrO_3, is calculated as follows:

$$\frac{(A-B) \times N \times 3.33\frac{1}{3}}{C} = \text{percent } CrO_3$$

where:

A = ml of $KMnO_4$ required for blank
B = ml of $KMnO_4$ required for sample
N = normality of $KMnO_4$
C = weight of sample

4.4.6 *Solubility.*—Accurately weigh a 3-gram sample into a 250-ml. beaker. Add 100 ml. of distilled water and heat to boiling. Boil for 10 minutes, replacing evaporated water from time to time. Filter the hot solution through a tared, asbestos-felted Gooch crucible. Wash the residue thoroughly with hot distilled water. Dry the crucible in an oven at 100°–105° C. for 3 hours, cool, and weigh. Calculate the gain in weight of the crucible as matter insoluble in water.

4.4.7 *Corrosion.*—Degrease one tinned steel strip, 2½ inches by ⅝-inch, conforming to type II, class A, grade 1 (95 pounds) weight per base box of Federal Specification QQ-I-706, by swabbing in a mixture of equal parts of alcohol and acetone. Allow the solvent to evaporate, heat in an oven at 105° C. for 30 minutes, cool and accurately weigh the strip. Completely immerse the strip in 200 ml. of a 3-percent distilled water solution of the sample contained in a 250-ml. beaker. Boil the solution for 1 hour, replacing evaporated water from time to time. At the end of 1 hour, remove the strip, rinse thoroughly with distilled water, then with acetone and dry in an oven at 105° C. for 30 minutes. Cool, reweigh the strip and examine for evidence of corrosion such as etching, discoloration, or spangling of the tin. Report change in weight of the strip in milligrams. Perform this test in duplicate.

4.4.8 *Coarse particles.*—Determine percent retained on a No. 8 Standard sieve conforming to Federal Specification RR-S-366 by method 2310.2 of Federal Specification P-S-536.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing for domestic shipment.—Unless otherwise specified, cleaning compound shall be packed in fiberboard or plywood drums, or in slack barrels, with fully removable tops, constructed in accordance with Consolidated Freight Classification requirements, rule 41 or in multiwall sacks (see 5.1.1). Not more than 125 pounds of the product shall be packed per container. Each container except multiwall sacks shall be lined with a bag liner of kraft paper of a minimum 45 pounds basis weight (24 x 36–500) creped so as to have a minimum stretch of 15 percent in both directions. Bags shall conform to the size of the container and be of sufficient depth to provide a minimum overlap of 3 inches when the mouth of the bag is closed and to allow slight movement of contents during transit without placing excessive strain on the bag liner. The mouth of the bag shall be securely closed by tying or sewing to prevent sifting of contents during transit.

5.1.1 *Construction of multiwall sacks.*—Construction of multiwall sacks shall conform to 4x, 9x, or 14x, of Federal Specification UU–S–48. Net weight shall not exceed 50 pounds.

6. NOTES

6.1 Ordering data.—Purchasers should specify the following:

(a) Title, number, and date of this specification.

(b) That baking pan cleaner compound shall be purchased on the basis of net weight.

(c) Whether preparation for domestic or overseas shipment is required (see sec. 5 and 7.2.2).

6.2 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated in the titles of the specifications, or which are commercially available but are intended to cover the types, etc., which are suitable for Federal Government requirements.

6.3 Transportation description.

6.3.1 *Transportation description applicable to this item is:*

Clearing, scouring, or washing compound not otherwise indexed by name:

Carload weight 36,000 pounds.
Truckload weight 36,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, standard, and publication of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 Army, Navy, and Air Force.

Federal Specification:

TT–E–485—Enamel; for Drums and Other Metal Products, Rust Inhibiting, Olive Drab.

Military Specifications:

JAN–P–100—Packaging and Packing for Overseas Shipment; General Specification.

JAN–D–111—Drums, Fiberboard (Overseas Type).

JAN–P–112—Packaging and Packing for Overseas Shipment; Drums, Plywood.

Military Standard:

MIL–STD–129—Marking of Shipments.

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

7.2.1 *Other publications.*—The following publication of the issue in effect on date of invitation for bids forms a part of this specification.

Interstate Commerce Commission Regulations:

Regulations for Transportation of Explosives and Other Dangerous Articles, Etc.

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(Application for copies of the above-named publication should be addressed to the Interstate Commerce Commission, Washington 25, D. C.)

7.2.2 *Packaging and packing for domestic shipment and storage or overseas shipment.*

7.2.2.1 Cleaning compound shall be packed in fiberboard or plywood drums of not more than 125 pound capacity, constructed in accordance with Military Specifications JAN-D-111 and JAN-P-112, respectively.

7.2.2.2 Alternatively, cleaning compound may be packed in metal drums of not more than 125 pound capacity, full opening lug cover type, made from not less than 22 gage bonderized steel throughout. Each container shall have one top and one bottom expanded bead or rolling hoop. The bottom chime shall be compound-lined and double-seamed to container body and side seam shall be welded. The full removable cover shall be made with not less than 16 lugs and shall be fitted with firmly affixed gasket of suitable composition. The containers shall be coated on the inside with an enamel suitable for the product and the exterior shall be coated overall with an enamel conforming to Federal Specification TT-E-485. The containers shall comply with the test requirements of Freight Classification No. 4 of Interstate Commerce Commission Regulation, specification for shipping containers, 37-D, Steel Drums.

7.2.2.3 *Liners.*—Each fiberboard or plywood drum shall be lined with a bag liner, constructed in accordance with grade 14 or 19 of Military Specification JAN-P-100. The mouth of the bag shall be securely closed by tying or sewing to prevent sifting of contents during transit. Metal drums will not require a bag liner.

7.2.3 *Marking.*

7.2.3.1 *Marking instructions.*—One instruction for use sheet, inclosed in an envelope, shall be packed in the top of each container in such a manner that sheet may be taken from container after cover is removed. (Instruction sheet shall also include warning against (a) use of aluminum pans (b) danger due to contact with human skin.)

7.2.3.2 *Shipping containers.*—Each shipping container shall be marked in accordance with Military Standard MIL-STD-129. The nomenclature shall be as follows:

Cleaning Compound, Baking Pan

MILITARY INTEREST:

Army—Q
Navy—Sh S Y
Air Force.

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